

Q.I) Choose the right answers and label them in your answer sheet. (50x1)

1. is a technical expert who integrates knowledge into a computer system when developing a decision support system, to solve complex problems that require human expertise.
A) domain expert B) knowledge analyzer C) knowledge designer ☒ D) Knowledge engineer
2. DSS supports all organization levels except..... level.
A) Strategic ☒ B) Operational C) Tactical D) Management
3. is an interactive graphical interface that makes the interaction easier between the DSS and its users
A) DSS SW system B) DSS DB ☒ C) DSS UI D) DSS KBS
4. are a system that employs knowledge about its application domain and uses an inferencing (reason) procedure to solve problems that would otherwise require human competence or expertise.
A) Decision support ☒ B) Expert systems C) Genetics D) Fuzzy systems
5. is a cognitive process that results in the selection of a course of action among several alternative scenarios
A) OALP B) Business analysis ☒ C) Decision making D) DSS
6. helps you to prioritize and identify the causes with the highest effect.
A) DT B) ERD ☒ C) Pareto chart D) Cause and effect
7. models are used to provide answers to what-if situations occurring frequently in an organization
☒ A) Sensitivity analysis B) Statistical C) Forecasting D) Goal seeking
8. supports decision-making by analyzing a given time-series of data and returning new information gained by those analysis
A) Model-Driven DSS B) Knowledge-driven DSS C) Document-driven DSS ☒ D) Data-driven DSS
9. Phase of the DSS process consists of choosing among solution alternatives
A) Intelligence B) Implementation ☒ C) Choice D) Design
10. Only part of the problem has a clear-cut answer provided by an accepted procedure in....decisions
A) Structured decisions ☒ B) Semi-structured C) Unstructured D) None of them
11. Decision-making under is assumed that complete knowledge is not available so that the decision-maker does not know what the outcome of each course of action will be.
☒ A) Certainty B) Risk ☒ C) Uncertainty D) Compress
12. provides an upper bound on the expected value of any sample or survey information.
A) AHP B) EV C) ROMC ☒ D) EVPI
13. The columns of the payoff table correspond to the
☒ A) States of nature B) Assumptions C) Alternatives D) Payoffs
14. is a financial tool that helps businesses determine the sales volume or revenue needed to cover all costs associated with producing and selling a product or service.
A) AHP model B) Pro forma ☒ C) Break even D) Ratio
15. is difficult to use when the number of alternatives is high.
A) ANN ☒ B) AHP C) DT D) EVP
16. In AHP, If the CR is the judgments are untrustworthy because they are too close comfort to randomness and the exercise is valueless or must be repeated.
A) 0.01 B) 0.005 C) 0.09 ☒ D) 0.5
17. In the influence diagram, represented by squares and represent the decisions need to be made
☒ A) decision nodes B) value nodes C) Chance nodes D) Output nodes

18.emphasizes the quantification of individual preferences using utility functions
 A) AHP ☒ B) MAUA C) DT D) EVP
19. based on the perceptions and opinions of experts instead of hard data, long-term forecasting.
 A) Moving average B) Regression ☒ C) Judgement methods ☒ D) prediction
20. may work either in batch mode, processing accumulated transactions at a single time later on, or in online mode, processing incoming transactions immediately.
☒ A) TPS B) MIS C) DSS D) GIS
21. You can use Cause-and-Effect diagrams and the Pareto Chart tool for in DSS process
 A) ☒ Information Gathering ☒ B) Identification of the Purpose of the Decision ☒ C) Brainstorm and Analyse the Choices ☒ D) Evaluation of Alternatives
22. is an organized approach to designing specialized decision support systems, more precisely their user interfaces.
 A) MAUA B) AHP C) GUI ☒ D) ROMC
23. is a model-driven DSS to help a person decide how much he or she needs to invest each month for retirement
☒ A) Fidelity B) Stock finder C) Microsoft CarPoint D) TCB Works
24. During phase, you conduct a thorough analysis of the current decision-making processes within the organization.
 A) Feasibility study B) Initialization ☒ C) Diagnosis D) Make/Buy
25. is a data-driven approach. We start from available information as it becomes available or from a basic idea, and then we try to draw conclusions
☒ A) Forward chaining B) Backward chaining C) Reinforcement D) Supervised DS
26. is a structured approach for evaluating and assessing the effectiveness, efficiency, and security of a Decision Support System within an organization.
☒ A) DSS audit plan B) DSS diagnosis C) DSS planning D) DSS evaluation
27. is an application of expert systems for systems configuration.
 A) MYCIN B) LMOS ☒ C) XCON D) SONAR
28. developed at Stanford University in the early 1980s for medical diagnosis, is the most well-known ES application
 A) SONAR B) LMOS C) XCON ☒ D) MYCIN
29. is an area in ES used for recording intermediate results, hypotheses, and decisions
☒ A) User interface ☒ B) Knowledge base ☒ C) Blackboard ☒ D) Database
30. The collection of intensive activities encompassing the acquisition of knowledge from human experts and conversion of this knowledge into a repository is called knowledge.....
 A) acquisition ☒ B) engineering C) evaluation D) validation
31. is the process of using the rules in the knowledge base along with the known facts to conclude.
 A) Expert systems B) DSS SW C) DSS ☒ D) Inferencing (or reasoning)
32. Random changes are applied to some individuals in the population in phase of the genetic algorithm phase.
 A) Crossover B) Termination C) Replacement ☒ D) Mutation
33. The fewer resources a system consumes in producing given outputs, the more it is.
☒ A) Efficient B) Reliable C) Secured D) Effective
34. Decisions are those in which the decision maker must provide judgment, evaluation, and insights into the problem definition
 A) Structured B) Semi-Structured ☒ C) Unstructured D) None of them
35. measuring performance against the planned objectives and initiating corrective action, if needed.
 A) Planning B) Leadership ☒ C) Controlling D) Organizing

*So The optimal car is (Mazda Miata)
 Criteria Weight style, Reliability, Fuel $\Rightarrow (0,30,0,60,0,10)$

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36. The decision-making process can be defined as the activity that leads to the solution of a decision-making problem involving at most Alternatives.
 A) One ☒ B) Two C) Three ☐ D) None of them
37. is a phase of the DSS process that consists of discovering, identifying, and understanding the problems occurring in the organization.
☒ A) Intelligence B) Design C) Control ☐ D) Implementation
38. The output of the DSS design phase is set of
 A) Problem statements B) Solutions ☒ C) Alternatives ☐ D) States of nature
39. If a company has fixed costs of \$100,000, variable costs of \$1000 per unit, and a selling price of \$2000 per unit, this means that the company must sell units to cover all its costs (BEA).
 A) 1000 ☐ B) 2000 ☐ C) 5000 ☒ D) 100
40. analysis can summarize the anticipated financial results for a specific period in the future
 A) AHP model ☒ B) Pro forma C) Break even ☐ D) Ratio
41. The process of is based on transforming data to information, then to decisions, and finally to actions.
☒ A) BI B) DSS C) Data mining ☐ D) Simon
42. Data availability is not available in which of the following levels?
☒ A) Strategic B) Operational C) Tactical ☐ D) Management
43. is an application of expert systems for stock market monitoring.
 A) MYCIN ☐ B) LMOS ☐ C) XCON ☒ D) SONAR
44. According to Table 2, The threshold value (0.5) sets the output Y to 1 in the next row if the weighted sum of inputs is greater than 0.5; otherwise, Y is set to 0, Alpha = 0.2, initially $w_1=0.1$, $w_2=0.3$, so the new weights after the first step are
 A) $w_1=0.1$, $w_2=0.3$ ☐ B) $w_1=0.1$, $w_2=0.5$ ☐ C) $w_1=0.3$, $w_2=0.3$ ☒ D) $w_1=0.3$, $w_2=0.5$
45. According to Table 1, the EVPI equals ($p(s_1)=0.5$, $p(s_2)=0.2$, $p(s_3)=0.3$).
 A) 9 ☐ B) 14.5 ☐ C) 1.25 ☒ D) 3.1
46. According to Table 1, the decision maker will choose, based on an optimistic approach. $\leftarrow$
☒ A) A1 B) A2 C) A3 ☐ D) A4
47. According to Table 1, the decision maker will choose, based on the Expected value approach. ($p(s_1)=0.5$, $p(s_2)=0.2$, $p(s_3)=0.3$).
☒ A) A1 B) A2 C) A3 ☐ D) A4
48. According to Table 1, the decision maker will choose, based on a pessimistic approach.
 A) A1 ☒ B) A2 C) A3 ☐ D) A4
49. According to Table 1, the decision maker will achieve, based on the Expected value approach. ($p(s_1)=0.5$, $p(s_2)=0.2$, $p(s_3)=0.3$).
 A) 1.75 ☐ B) 3.6 ☒ C) 13.4 ☐ D) 14.5
50. According to Table 1, the decision maker will choose, based on the regret approach.
☒ A) A1 B) A2 C) A3 ☐ D) A4

Q.2) Answer the following question (10 marks).

According to Figure 1, use the following problem description:

- Objective: Selecting a car
 - Criteria: Style, Reliability, Fuel-economy?
 - Alternatives: Civic Coupe, Saturn Coupe, Ford Escort, Mazda Miata
- Using the AHP method, write all the required remaining steps to select the optimal car.

$$16.5 - 13.4 = 3.1$$